

CSS426 Machine Learning Operations (MLOps) Batch1

Lab_5

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Task: Self-Hosting and Configuring Jenkins Using Docker

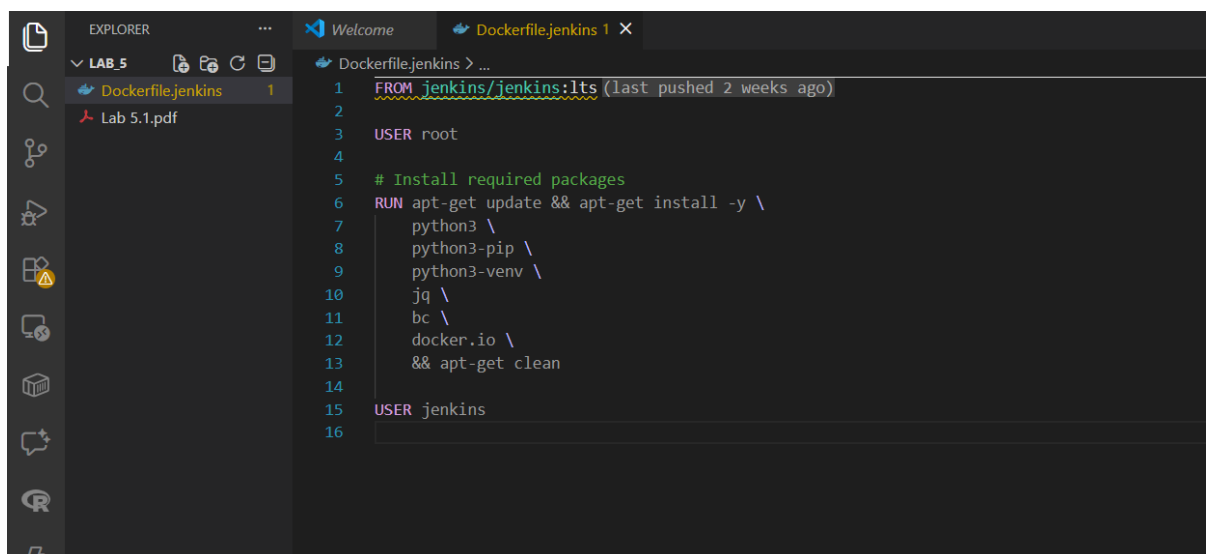
Objective:

This lab introduces students to setting up Jenkins as a self-hosted CI/CD server using Docker. Students will install, run, and configure Jenkins locally, preparing it for use in subsequent MLOps labs.

Task 1: Create the Custom Jenkins Docker Image

```
D:\SEM-VIII\CSS426 Machine Learning Operations(MLOps)\Lab_5>docker --version
Docker version 27.4.0, build bde2b89
```

```
D:\SEM-VIII\CSS426 Machine Learning Operations(MLOps)\Lab_5>|
```



```
1 FROM jenkins/jenkins:lts (last pushed 2 weeks ago)
2
3 USER root
4
5 # Install required packages
6 RUN apt-get update && apt-get install -y \
7     python3 \
8     python3-pip \
9     python3-venv \
10    jq \
11    bc \
12    docker.io \
13    && apt-get clean
14
15 USER jenkins
16
```

→ Dockerfile.jenkins is created

```
D:\SEM-VIII\CSS426 Machine Learning Operations(MLOps)\Lab_5>docker build -t 2022bcd0018-jenkins -f Dockerfile.jenkins .
[+] Building 268.4s (6/6) FINISHED
=> [internal] load build definition from Dockerfile.jenkins
=> => transferring dockerfile: 290B
=> [internal] load metadata for docker.io/jenkins/jenkins:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/2] FROM docker.io/jenkins/jenkins:latest@sha256:d1ea795c6facd7f549a21c40e5e43ffcc5fbc5f48683d9b24750f26e8079d772
=> => resolve docker.io/jenkins/jenkins:latest@sha256:d1ea795c6facd7f549a21c40e5e43ffcc5fbc5f48683d9b24750f26e8079d772
=> => sha256:7ad60f966cc569b8cd2fb9743561aefc702562607734da24feed2fe0ae60e6c9 392B / 392B
=> => sha256:57364cf49183c5a348ee44f64d97824f3675ca755e443ac7428e9b0fe33c53e 1.29kB / 1.29kB
=> => sha256:ed559592686a018bd4133dbddefe59a393c76879d7de4898839197efee53cf85 2.23kB / 2.23kB
=> => sha256:99ddb13f3ebf7c016ff3c4e840f7f94e8ae2ee11e76ddd68473ec41c42da5be7 72.05MB / 72.05MB
=> => sha256:39483126681f342b217818e59842ad78a21576eec7da2432104f8362d4291b1e 6.29MB / 6.29MB
=> => sha256:406f94702a48c7fc4bd96e198cace8344da8ad779b02cc0daba4a09bf150b8fc 189B / 189B
=> => sha256:e92c5cf141dc90da610685a63439962c05cd31a20c926e06f3289c0705c3a599 96.12MB / 96.12MB
=> => sha256:213d96e0ed060c4bce7c21be00d4ee2bc539c70ecce57c801ed206202eabeaf1 182B / 182B
=> => sha256:5b302cbb8ad382255e4945cbb4b538a40d327abab74e027a7f4f4f431d174e94 1.23kB / 1.23kB
=> => sha256:alc44ba8dd7aa3d28eb657a8b982f61639d88be1b791be797bcf4bb8208f8ce7a 5.24MB / 5.24MB
=> => sha256:17e22d939f1732029823f86562d22d277dd321da6f370e2126eeb2577a2d49af 61.59MB / 61.59MB
=> => sha256:53c88f1dfef79b2f207f7f1a03a45e0dc5ed208b9f496de16b98f81189dc0392 49.29MB / 49.29MB
=> => extracting sha256:53c88f1dfef79b2f207f7f1a03a45e0dc5ed208b9f496de16b98f81189dc0392 1.9s
=> => extracting sha256:17e22d939f1732029823f86562d22d277dd321da6f370e2126eeb2577a2d49af 2.2s
=> => extracting sha256:alc44ba8dd7aa3d28eb657a8b982f61639d88be1b791be797bcf4bb8208f8ce7a 0.2s
=> => extracting sha256:5b302cbb8ad382255e4945cbb4b538a40d327abab74e027a7f4f4f431d174e94 0.0s
=> => extracting sha256:213d96e0ed060c4bce7c21be00d4ee2bc539c70ecce57c801ed206202eabeaf1 0.0s
=> => extracting sha256:e92c5cf141dc90da610685a63439962c05cd31a20c926e06f3289c0705c3a599 0.4s
=> => extracting sha256:406f94702a48c7fc4bd96e198cace8344da8ad779b02cc0daba4a09bf150b8fc 0.0s
=> => extracting sha256:39483126681f342b217818e59842ad78a21576eec7da2432104f8362d4291b1e 0.1s
=> => extracting sha256:99ddb13f3ebf7c016ff3c4e840f7f94e8ae2ee11e76ddd68473ec41c42da5be7 0.7s
=> => extracting sha256:ed559592686a018bd4133dbddefe59a393c76879d7de4898839197efee53cf85 0.0s
=> => extracting sha256:57364cf49183c5a348ee44f64d97824f3675ca755e443ac7428e9b0fe33c53e 0.0s
```

```

=> => sha256:17e22d939f1732029823f86562d22d277dd321da6f370e2126eeb2577a2d49af 61.59MB / 61.59MB
=> => sha256:53c88f1dfef79b2f207f7f1a03a45e0dc5ed208b9f496de16b98f81189dc0392 49.29MB / 49.29MB
=> => extracting sha256:53c88f1dfef79b2f207f7f1a03a45e0dc5ed208b9f496de16b98f81189dc0392 1.9s
=> => extracting sha256:17e22d939f1732029823f86562d22d277dd321da6f370e2126eeb2577a2d49af 2.2s
=> => extracting sha256:alc44ba8dd7aa3d28eb657a8b982f61639d88be1b791be797bcf4bb8208f8ce7a 0.2s
=> => extracting sha256:5b302cbb8ad382255e4945cbb4b538a40d327abab74e027a7f4f4f431d174e94 0.0s
=> => extracting sha256:213d96e0ed060c4bce7c21be00d4ee2bc539c70ecce57c801ed206202eabeaf1 0.0s
=> => extracting sha256:e92c5cf141dc90da610685a63439962c05cd31a20c926e06f3289c0705c3a599 0.4s
=> => extracting sha256:406f94702a48c7fc4bd96e198cace8344da8ad779b02cc0daba4a09bf150b8fc 0.0s
=> => extracting sha256:39483126681f342b217818e59842ad78a21576eec7da2432104f8362d4291b1e 0.1s
=> => extracting sha256:99ddb13f3ebf7c016ff3c4e840f7f94e8ae2ee11e76ddd68473ec41c42da5be7 0.7s
=> => extracting sha256:ed559592686a018bd4133dbddefe59a393c76879d7de4898839197efee53cf85 0.0s
=> => extracting sha256:57364cf49183c5a348ee44f64d97824f3675ca755e443ac7428e9b0fe33c53e 0.0s
=> => extracting sha256:7ad60f966cc569b8cd2fb9743561aefc702562607734da24feed2fe0ae60e6c9 0.0s
=> [2/2] RUN apt-get update && apt-get install -y python3 python3-pip python3-venv jq bc docker.io
=> => exporting to image
=> => exporting layers
=> => exporting manifest sha256:9d7cb4be274aa239e38aa07367e5aaa76308872db9061d98978f26151a7b016f
=> => exporting config sha256:e51272769644549b5a91ce825b91fb05a7f1cc5765f5c80636f512dfbb159374
=> => exporting attestation manifest sha256:e05cf3c8cdf05aff7328a6bc42eeae1f7c9aeb94a3c0927e740961c84940d
=> => exporting manifest list sha256:5e69ca03d87874ca39e3067b503bd23c5b29bbe512c36ee2e92e95f69e1a53bc
=> => naming to docker.io/library/2022bcd0018-jenkins:latest
=> => unpacking to docker.io/library/2022bcd0018-jenkins:latest

```

D:\SEM-VIII\CSS426 Machine Learning Operations(MLOps)\Lab_5>docker images

→ Built the custom Jenkins image using Docker.

```
D:\SEM-VIII\CSS426 Machine Learning Operations(MLOps)\Lab_5>docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
2022bcd0018-jenkins latest             5e69ca03d878       About a minute ago 1.93GB
druva21/wine-inference latest             2c96777527328      2 weeks ago        686MB
<none>              <none>            <none>              <none>              <none>
druva21/druva_2022bcd0018_wine v1                 95300d6d49a1       3 weeks ago        580MB
2022bcd0018-druvakumar latest             58306b784979       10 months ago      248MB
2022bcd0018-druva   latest             b80ac8264375       10 months ago      223MB
sample              latest             5ca2452af357       10 months ago      131MB
my-webapp           latest             9e79371456db       10 months ago      785MB
sonarqube           latest             c0a734bd7e62       11 months ago      2.12GB
microservices-app-worker latest             3db0471290a5       11 months ago      1.49GB
microservices-app-web latest             8fca4df540dc       11 months ago      1.47GB
2022bcd0018_web     latest             90048594a556       11 months ago      1.47GB
18_1                latest             38c1b30618e6       11 months ago      1.47GB
postgres            15                 e897e4ad3501       11 months ago      608MB
2022bcd0018         latest             ad73e3e8772eb      12 months ago      1.47GB
todo-app            latest             ae8906a9191f       12 months ago      409MB
postgres            latest             3267c505060a       12 months ago      614MB
hello-world         latest             d715f14f9eca       12 months ago      20.4kB
gcr.io/k8s-minikube/kicbase v0.0.46           fd2d445ddcc3       12 months ago      1.86GB
gcr.io/k8s-minikube/kicbase <none>            fd2d445ddcc3       12 months ago      1.86GB
nginx                latest             0a399eb16751       14 months ago      278MB
nareshm988/firstrepo latest             9c1c841d971b       14 months ago      117MB
ubuntu              latest             80dd3c3b9c6c       14 months ago      117MB
maven                3.9.6-eclipse-temurin-17 29a1658b1f30       2 years ago        728MB
confluentinc/cp-kafka 7.4.1              d3acff0466ec       2 years ago        1.33GB
confluentinc/cp-zookeeper 7.4.1              c642265b70fb       2 years ago        1.33GB
confluentinc/cp-kafka 7.4.0              187dac6627e7       2 years ago        1.33GB
confluentinc/cp-zookeeper 7.4.0              90631f224b43       2 years ago        1.33GB
```

→ At the top, we are clearly seeing the created image.

Task 2: Run Jenkins Using the Custom Image

```
D:\SEM-VIII\CSS426 Machine Learning Operations(MLOps)\Lab_5>docker run -d -u root -p 9090:8080 -p 50000:50000 -v //var/run/docker.sock:/var/run/docker.sock -v jenkins_home:/var/jenkins_home --name 2022bcd0018_jenkins 2022bcd0018-jenkins 1daf7269ce0334abef0a7b31245936329689a835f34618cbd292d31e0d4a5b4
```

```
D:\SEM-VIII\CSS426 Machine Learning Operations(MLOps)\Lab_5>docker ps
```

→Running the Jenkins using that custom image created.

Task 3: Verify Jenkins Container Status

```
D:\SEM-VIII\CSS426 Machine Learning Operations(MLOps)\Lab_5>docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
1daf7269ce03	2022bcd0018-jenkins	"/usr/bin/tini -- /u..."	12 seconds ago	Up 12 seconds	0.0.0.0:50000->50000/tcp, 0.0.0.0:9090->8080/tcp

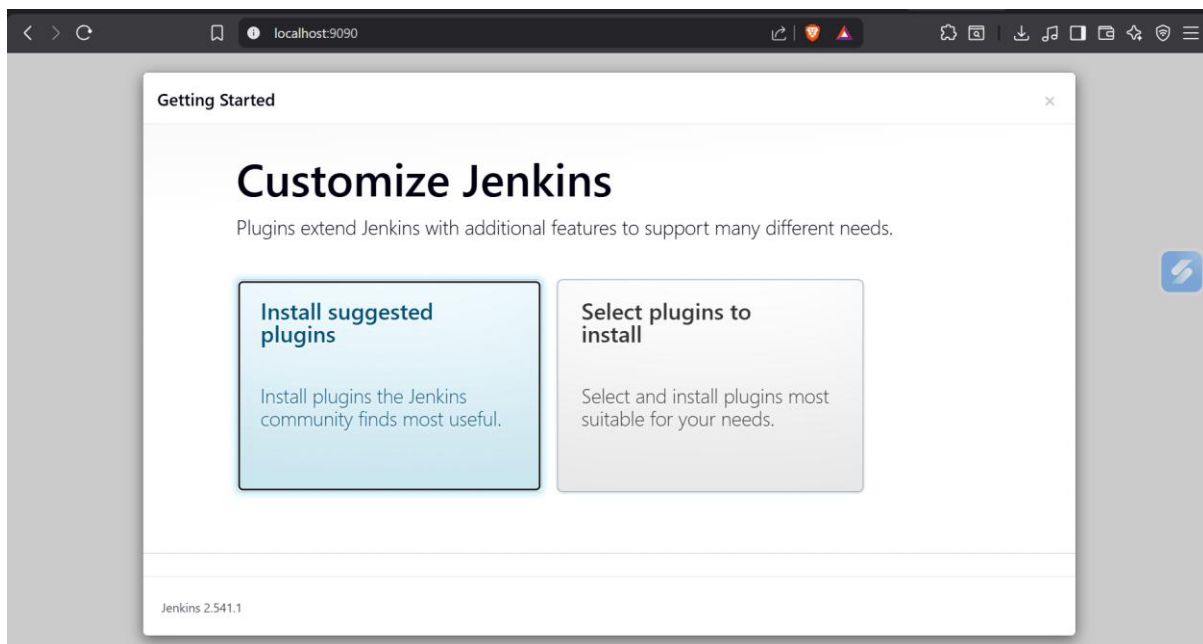
→From this image, we can see that Jenkins is running on port 9090 using that Jenkins image.

Task 4: Unlock Jenkins

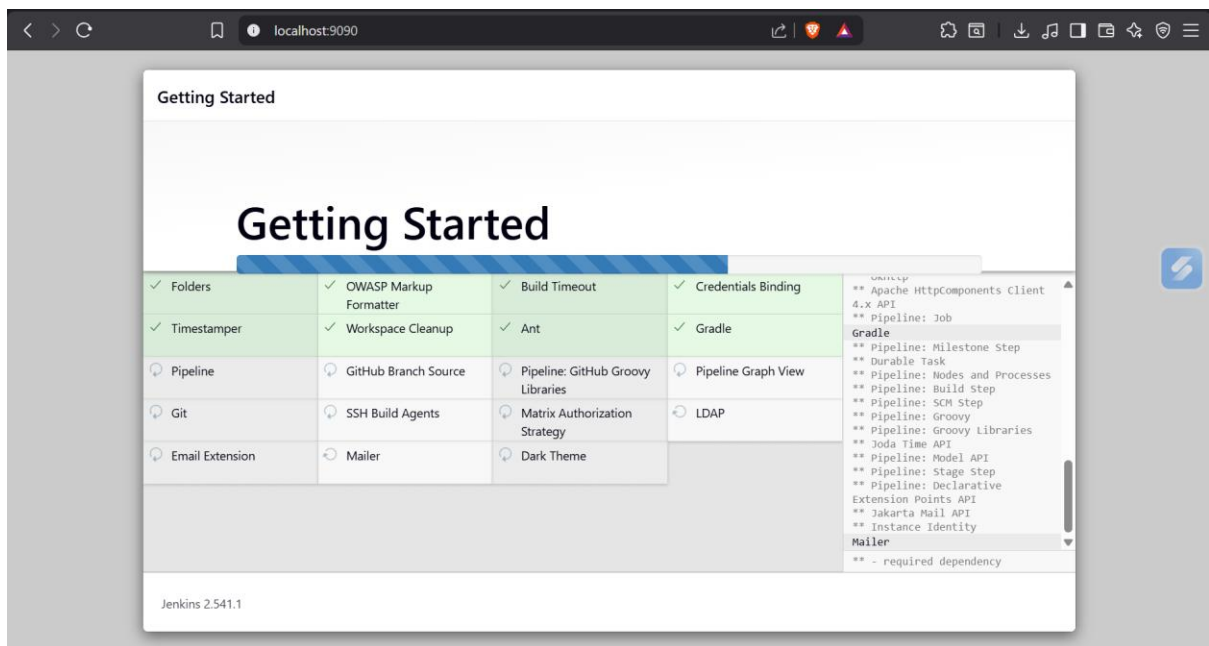
```
D:\SEM-VIII\CSS426 Machine Learning Operations(MLOps)\Lab_5>docker exec 2022bcd0018_jenkins cat /var/jenkins_home/secrets/initialAdminPassword
d50fc8ebb81b4fffa00a3c54b9141e14
```

```
D:\SEM-VIII\CSS426 Machine Learning Operations(MLOps)\Lab_5>docker images
```

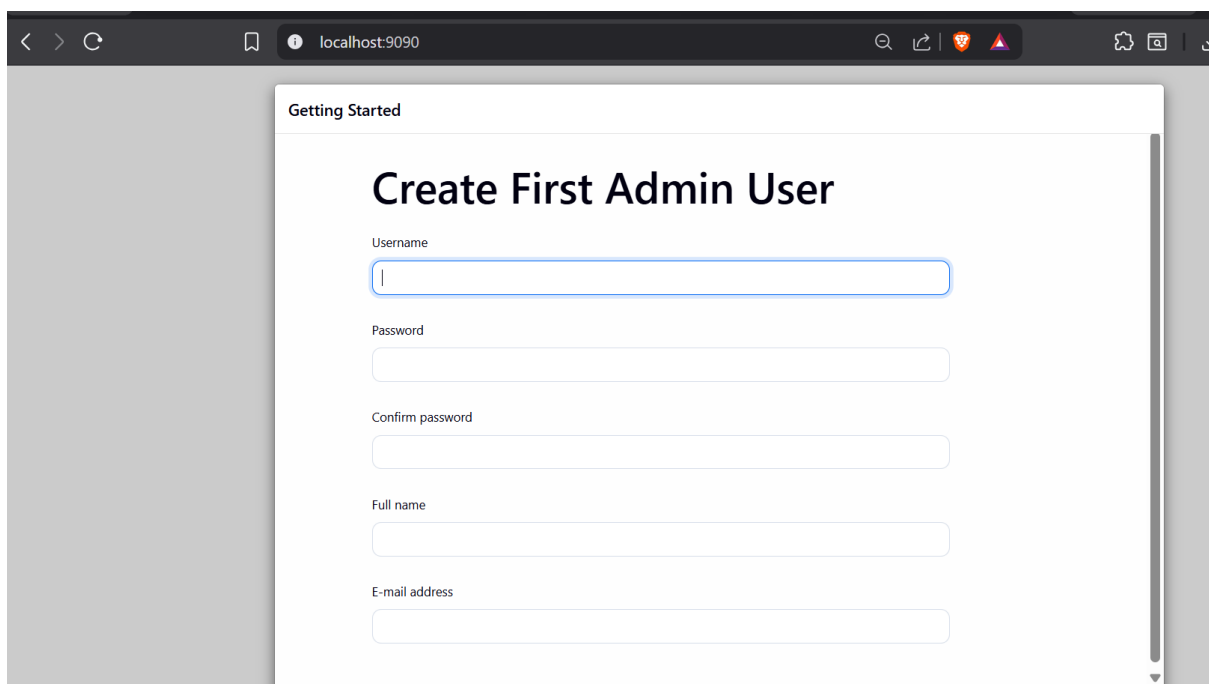
→ Retrieved the initial password to log in.

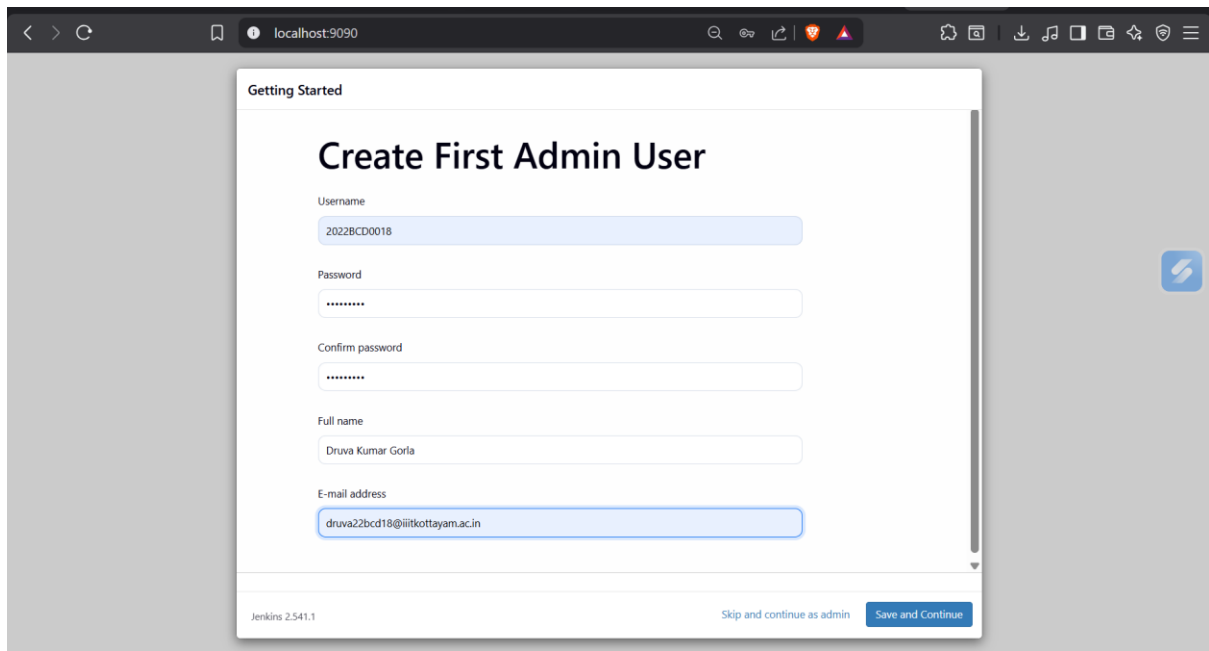


→On port 9090, Jenkins is running and we are installing suggested plugins.



→After installing the plugins, we have to create an Admin User.



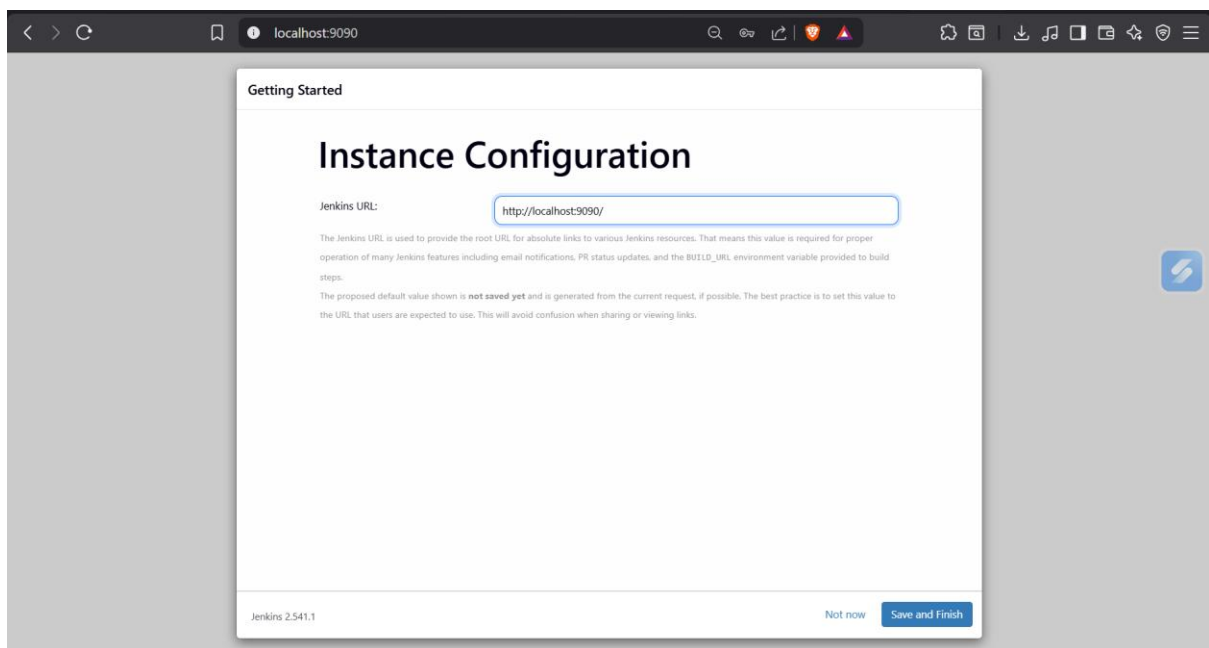


The screenshot shows the Jenkins 'Getting Started' page with the 'Create First Admin User' form. The form contains the following fields and values:

Field	Value
Username	2022BCD0018
Password	*****
Confirm password	*****
Full name	Druva Kumar Gorla
E-mail address	druva2bcd18@iiitkottayam.ac.in

At the bottom of the form, there are two buttons: 'Skip and continue as admin' and 'Save and Continue'. The Jenkins version 'Jenkins 2.541.1' is displayed in the bottom left corner.

→ Here, I have created an Admin using my details.

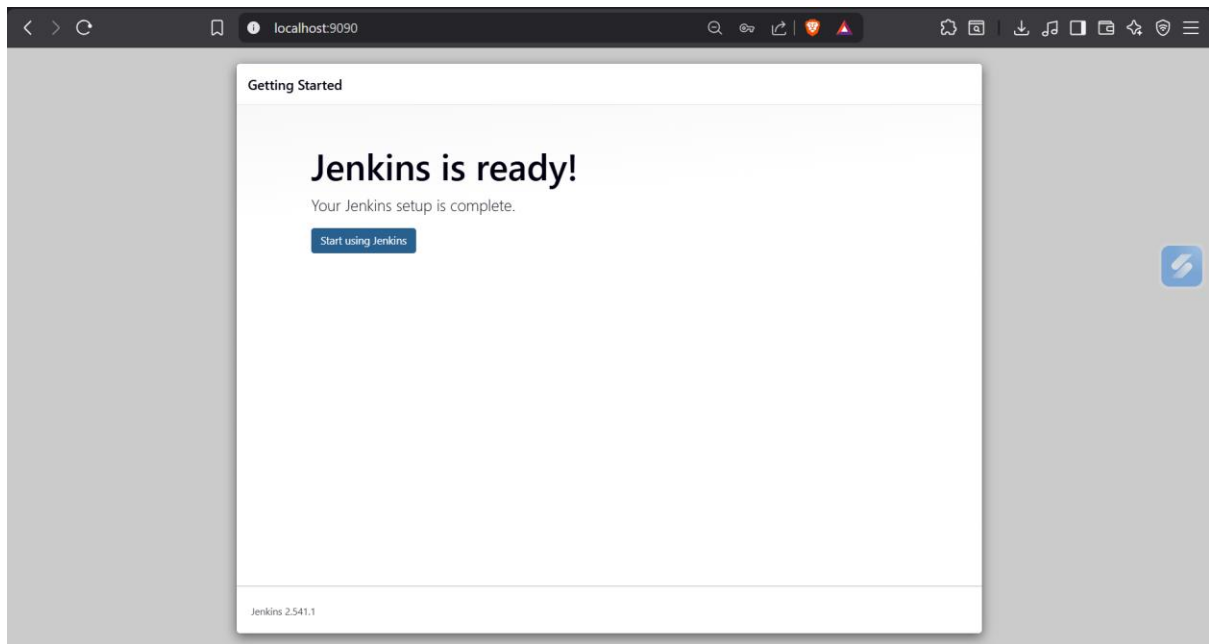


The screenshot shows the Jenkins 'Getting Started' page with the 'Instance Configuration' form. The form contains the following field and value:

Field	Value
Jenkins URL:	http://localhost:9090/

Below the field, there is explanatory text: 'The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps. The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.'

At the bottom of the form, there are two buttons: 'Not now' and 'Save and Finish'. The Jenkins version 'Jenkins 2.541.1' is displayed in the bottom left corner.



→ From this image, it is clear that:

- Jenkins is Running in 9090.
- Logged in as user Druva Kumar Gorla, with ID 2022BCD0018.
- Jenkins Dashboard.

